

1. The period of oscillation of an object in an ideal spring and mass system is 0.50 s and the amplitude is 5.0 cm. what is the speed at the equilibrium point? and the acceleration at the point of maximum extension of the spring. (62.8 cm/s 7.9 m/s²)

$$T = 0.50 \text{ sec}$$

$$\text{Amplitude} = x_0 = 5 \text{ cm} = 0.05 \text{ m}$$

$$\text{Speed at equilibrium} = v = ?$$

$$\text{acceleration at extreme position} = a = ?$$

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Sol:-

$$v = r\omega$$

$$v = x_0 \left(\frac{2\pi}{T} \right)$$

$$v = (0.05) \left(\frac{2(3.142)}{0.5} \right)$$

$$v = 0.628 \text{ m/s}$$

$$\therefore a = r\omega^2$$

$$a = (0.05) \left(\frac{2\pi}{T} \right)^2$$

$$a = (0.05) \left(\frac{2(3.142)}{0.5} \right)^2$$

$$a = 7.89 \text{ m/sec}^2$$

Q2

2. A sewing machine needle moves with a rapid vibratory motion, like SHM, as it sews a seam. Suppose the needle moves 8.4 mm from its highest to its lowest position and it makes 24 stitches in 9.0s. What is the maximum needle speed? (7.0 cm/s)

Data:-

$$2x_0 = 8.4 \text{ mm}$$

$$x_0 = 4.2 \text{ mm} = 4.22 \times 10^{-3} \text{ m}$$

number of stitches = 24

Time for 24 stitches = 9 sec

Max speed = $V_{\max} = ?$

Sol:-

$$T = \frac{9}{24} = 0.375$$

$$V = r\omega$$

$$V = r \left(\frac{2\pi}{T} \right)$$

$$V = 4.22 \times 10^{-3} \left(\frac{2\pi}{0.375} \right)$$

$$V = 0.0707 \text{ m/sec}$$

3. An ideal spring with a spring constant of 15 N/m is suspended vertically. A body of mass 0.60 kg is attached to the unstretched spring and released.

Q3

(a) What is the extension of the spring when the speed is a maximum?

(b) What is the maximum speed?

[(a) 0.39 m

(b) 2.0 m/s]

$F = w$

Data:-

$$K = 15 \text{ N/m} \quad V_{\max} = ?$$

$$m = 0.6 \text{ kg}$$

$$x_0 = ?$$

Sol:-

Since spring vibrates vertically,

$$F = w$$

$$Kx_0 = mg$$

$$x_0 = \frac{mg}{K}$$

$$x_0 = \frac{(0.6)(9.8)}{15} \quad x_0 = 0.3924$$

$$\begin{aligned}
 V_{\max} &= r\omega \\
 &= \eta_0 \sqrt{\frac{k}{m}} \\
 &= 0.392 \sqrt{\frac{15}{0.6}}
 \end{aligned}$$

$$V_{\max} = 1.96$$

Q 7

4. A body is suspended vertically from an ideal spring of spring constant 2.5 N/m. the spring is initially in its relaxed position. The body is then released and oscillates about its equilibrium position. The motion is described by $y = (4.0\text{cm}) \sin[(0.70\text{rad/s}) t]$. What is the maximum kinetic energy of the body? (2.0 mJ)

$$K = 2.5 \text{ N/m}$$

$$y = 4 \text{ cm} [(\sin 0.7 \text{ rad/s} \cdot t)]$$

Sol.

$$y = 4 [(\sin 0.7 \text{ rad/sec} \cdot t)] \rightarrow (i)$$

$$y = A \sin \theta$$

$$y = \eta \sin \omega t \rightarrow (ii)$$

comp eq (i) and (ii)

$$\omega = 0.7 \text{ rad/s}$$

$$\eta_0 = 4 \text{ cm}$$

$$\eta_0 = 0.04 \text{ m}$$

rough

$$y = A \sin \theta$$

$$y = \eta_0 \sin \omega t$$

$$\omega = \frac{\theta}{t}$$

$$\theta = \omega t$$

$$= K.E_{\text{max}} = \frac{1}{2} k x_0^2$$

$$\Rightarrow K.E = \frac{1}{2} (255)(0.04)^2$$

$$\Rightarrow K.E = 2 \times 10^{-3} \text{ J}$$

Q5

5. The period of oscillation of a simple pendulum does not depend on the mass of the bob. By contrast the period of a mass-spring system does depend on mass. Explain the apparent condition.

Since for simple pendulum

$$T = 2\pi \sqrt{\frac{L}{g}}$$

or

$$T \propto \sqrt{L}$$

Time period of simple pendulum is directly proportional to square root of length

for spring mass system;

$$T = 2\pi \sqrt{\frac{m}{k}}$$

or

$$T \propto \sqrt{m}$$

Time period of mass spring system is independent of the length

Q6

6. What is the period of a simple pendulum of a 6.0 kg mass oscillating on a 4.0 m long string? (4.01s)

Data:-

$$T = ?$$

$$m = 6 \text{ kg}$$

$$L = 4 \text{ m}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$T = 2\pi \sqrt{\frac{4}{9.8}}$$

$$T = 4.01 \text{ sec}$$

7. A pendulum of length 75 cm and mass 2.5 kg swings with a mechanical energy of 0.015 J. what is its amplitude? (3.0 cm)

Q7

$$L = 75 \text{ cm} = 0.75 \text{ m}$$

$$\text{mass} = m = 2.5 \text{ kg}$$

$$M.E = 0.015 \text{ J}$$

$$X_0 = \text{Amplitude} = ?$$

Sol:-

$$\therefore T = 2\pi \sqrt{\frac{L}{g}}$$

but,

$$T = \frac{2\pi}{\omega}$$

$$2\pi \sqrt{\frac{L}{g}} = \frac{2\pi}{\omega}$$

$$\sqrt{\frac{L}{g}} = \frac{1}{\omega}$$

$$\omega = \sqrt{\frac{g}{L}}$$

$$\sqrt{\frac{k}{m}} = \sqrt{\frac{g}{L}}$$

Squaring o.b.s

$$\frac{k}{m} = \frac{g}{L}$$

$$\cancel{\frac{k}{m}} = k = \frac{g \times m}{L}$$

$$K = \frac{9.8 \times 2.5}{0.75}$$

$$K = 32.66 \text{ N/m}$$

$$\therefore E = \frac{1}{2} k x_0^2$$

$$\frac{2E}{K} = x_0^2$$

$$N_0 = \frac{2 \times 0.015}{32.66}$$

$$N_0 = 0.03 \text{ m} \text{ or } 3 \text{ cm}$$

8. A pendulum of length L_1 has a period of $T_1 = 0.950 \text{ s}$. the length of the pendulum is adjusted to a new value L_2 such that $T_2 = 1.00 \text{ s}$ what is the ratio L_2/L_1 (1.11)

Q8 Length = L_1

$$\text{Time period} = T_1 = 0.95 \text{ s}$$

$$\text{Length} = L_2$$

$$\text{Time period} = T_2 = 1 \text{ s}$$

$$\frac{L_2}{L_1} = ?$$

Sol:-

$$\therefore T_1 = 2\pi \sqrt{\frac{L_1}{g}} \rightarrow (i)$$

$$\therefore T_2 = 2\pi \sqrt{\frac{L_2}{g}} \rightarrow (ii)$$

divide eq (ii) by (i)

$$\therefore \frac{T_2}{T_1} = \frac{2\pi \sqrt{\frac{L_2}{g}}}{2\pi \sqrt{\frac{L_1}{g}}}$$

$$\frac{T_2}{T_1} = \frac{\sqrt{L_2}}{\sqrt{L_1}}$$

squaring o.b.s

$$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{\sqrt{L_2}}{\sqrt{L_1}}\right)^2$$

$$\left(\frac{1}{0.95}\right)^2 = \frac{L_2}{L_1}$$

$$\frac{L_2}{L_1} = 1.11$$

9. A wire is hanging from the top of a tower such that the top is not visible due to darkness. How do you calculate the height of tower? *vimp*

Q9

1) Attach a body of 'm' to the wire so it becomes a pendulum. 2) Now, wire starts vibrating like a pendulum.

3) Then, find the time period of the pendulum.

4) By using this time period, calculate the length of wire which is equal to the height of the tower.

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$\frac{gT^2}{4\pi^2} = l$$

Q10. 10. The amplitude of oscillation of a pendulum decays by a factor of 20.0 in 120 s. By what factor has its energy decayed in that time

(The energy has decreased by a factor of 400)

Data:-

max Amplitude = X_0

Reduced Amplitude = $\frac{X_0}{20}$

Time = $t = 120 \text{ sec}$

Energy decay = $E' = ?$

Sol:-

$$E = \frac{1}{2} k X_0^2 \rightarrow (i)$$

$$E' = \frac{1}{2} k \left(\frac{X_0}{20} \right)^2$$

$$E' = \frac{1}{2} k X_0^2 \left(\frac{1}{400} \right)$$

$$E' = \frac{E}{400}$$

Energy has decreased by a factor of 400